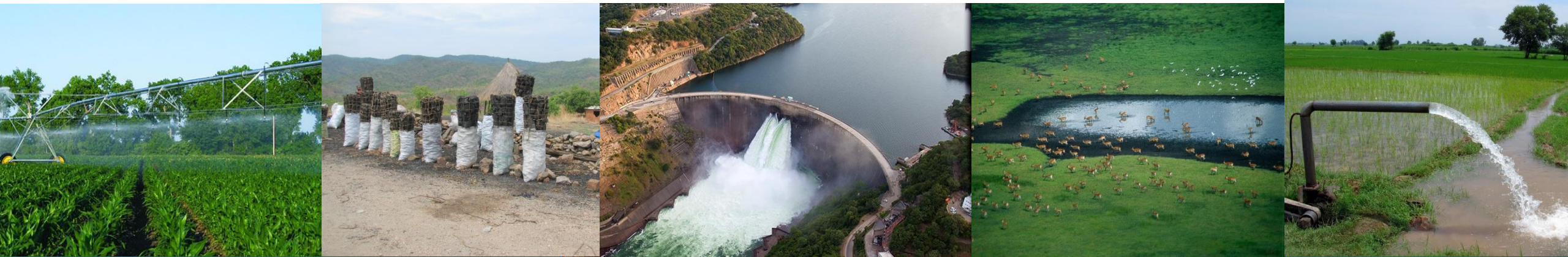


Exercise 3a: Calibration

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Water Security Group



Calibration

0. What you need
1. Introduction into calibration
3. A fast calibration
4. Visualization of the results



Calibration

0. What you need

- Python 3.7 - 3.12
- CWatM running as python version
- additional library: deap
 pip install deap (or conda install deap)
- and the library: matplotlib

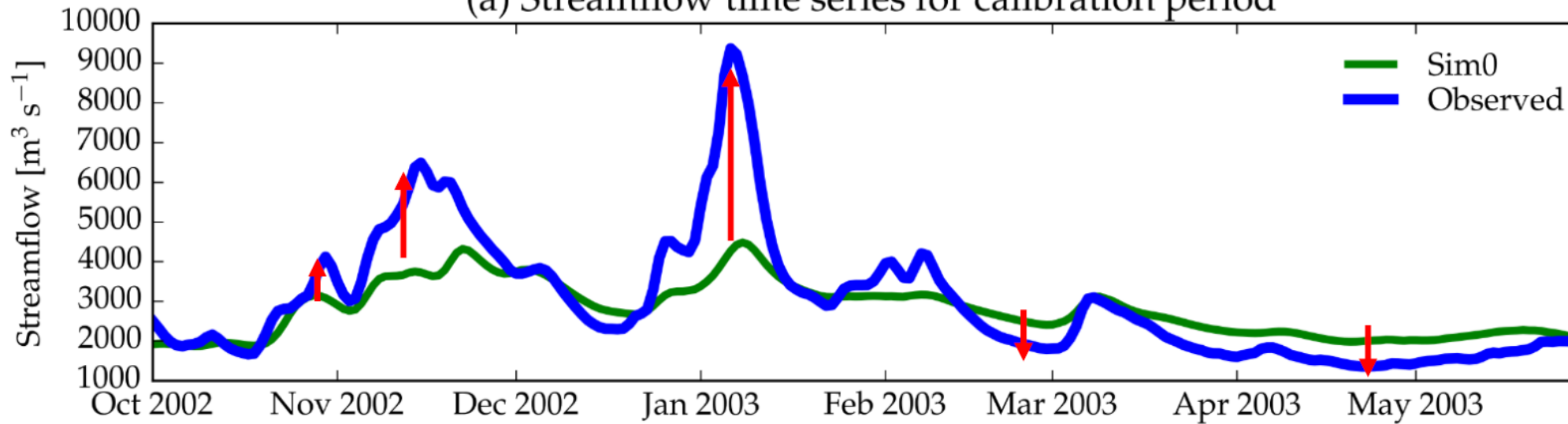


Calibration

Finding a parameter set which represents the observed discharge data



(a) Streamflow time series for calibration period



Calibration CWatM

Parameters

13 parameters are calibrated:

Snow

1. Snowfactor as a factor to amount of snow
2. SnowMelt: day-degree factor how much snow is melted per °C

Evapotranspiration

3. Crop factor as an adjustment to crop evapotranspiration

Soil

4. ksatfactor: a factor for the conductivity of soil of soil layer 1 and 2
5. soilfactor: a factor to the depth of soil layer 2
6. Preferential bypass flow: empirical shape parameter of the preferential flow relation
7. Infiltration capacity parameter: empirical shape parameter b of the ARNO model

Groundwater

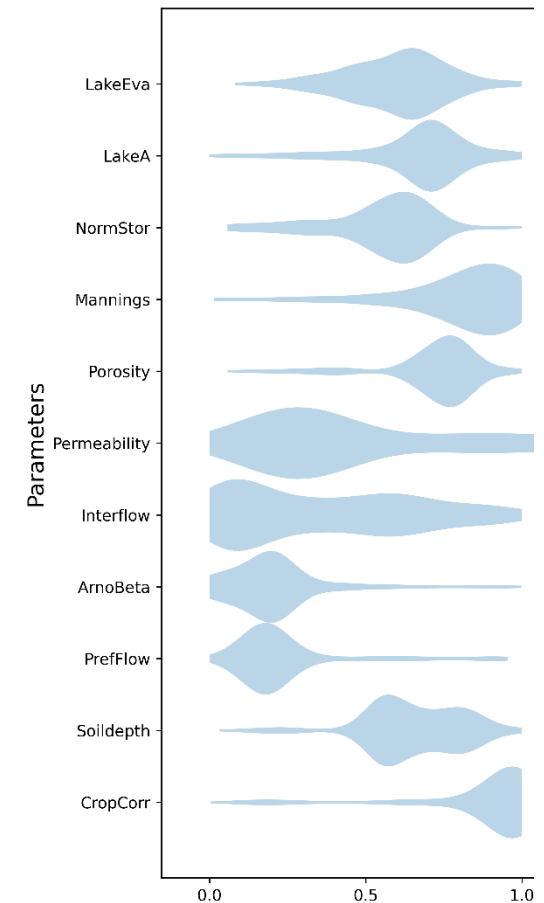
8. Interflow factor: factor to adjust the amount which percolates from interflow to groundwater
9. Recession coefficient: how fast is groundwater translated into baseflow

Routing

10. Runoff concentration: how fast is the runoff at the outlet of a grid cell
11. Channel Manning's n factor: a factor roughness in channel routing

Reservoir & lakes

12. Normal storage limit: the fraction of storage capacity used as normal storage limit
13. Lake A factor : factor to channel width and weir coefficient as a part of the Poleni weir equation



Parameter space for a calibration runs

Calibration CWatM

Objective function

Discharge:

Daily pairs of observed and simulated discharge at gauging stations

Objective function:

Modified version of the Kling-Gupta Efficiency (Kling et al., 2012),

$$KGE' = 1 - \sqrt{(r - 1)^2 + (\beta - 1)^2 + (\gamma - 1)^2}$$

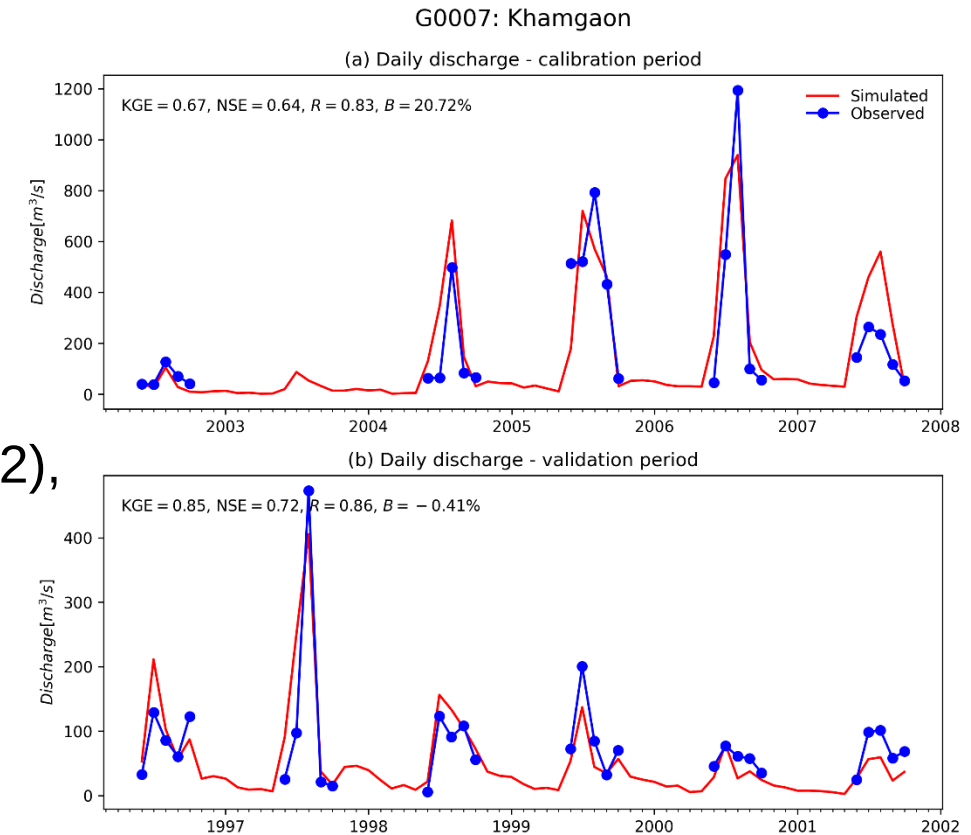
$$\beta = \frac{\mu_s}{\mu_o} \quad \text{and} \quad \gamma = \frac{CV_s}{CV_o} = \frac{\sigma_s/\mu_s}{\sigma_o/\mu_o}$$

r as the correlation coefficient between simulated and observed discharge (dimensionless),
 β as the bias ratio (dimensionless) and γ as the variability ratio.

CV is the coefficient of variation, μ is the mean streamflow [$\text{m}^3 \text{s}^{-1}$]

σ is the standard deviation of the streamflow [$\text{m}^3 \text{s}^{-1}$].

KGE' , r , β and γ have their optimum at 1.



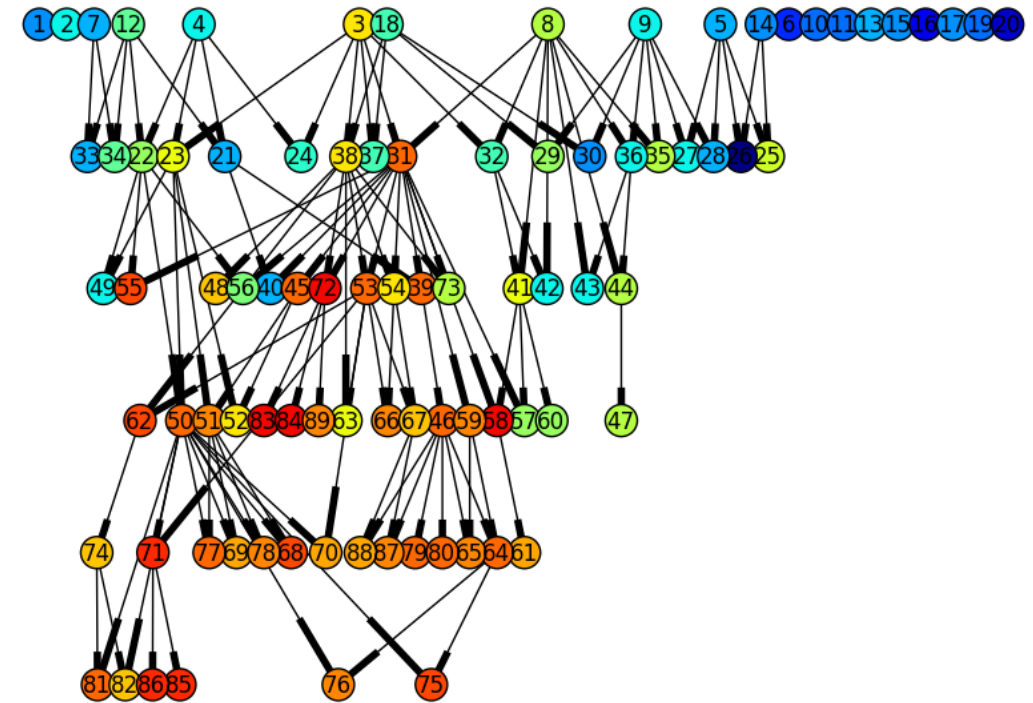
Calibration CWatM on discharge

Method

Calibration is using an evolutionary computation framework in Python called DEAP (Fortin et al., 2012).

DEAP implemented the evolutionary algorithm NSGA-II (Deb et al., 2002) which is used here as single objective optimization.

- Calibration start with an initial population size (μ)
- Each generation the best pool (λ) is selected and recombined
- After a number of generation, the best fitting calibration is found

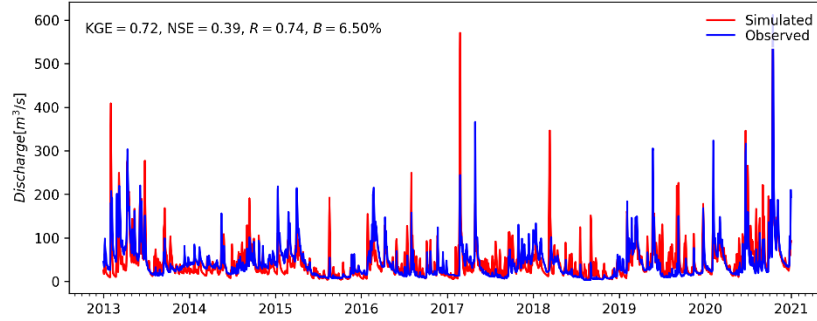


Evolution of parameter space

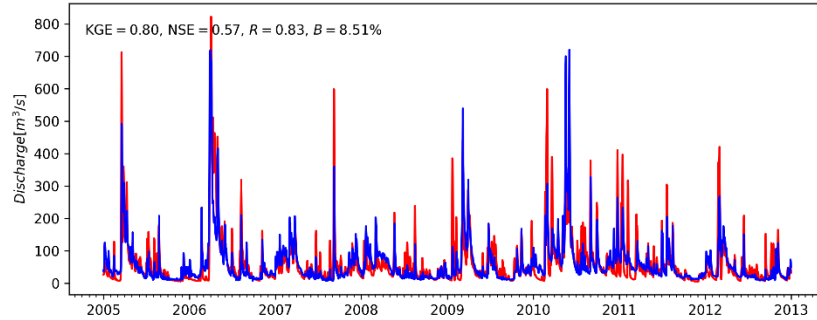
Calibration

MORAVA at STRAZNICE, basin area: 9145 km²

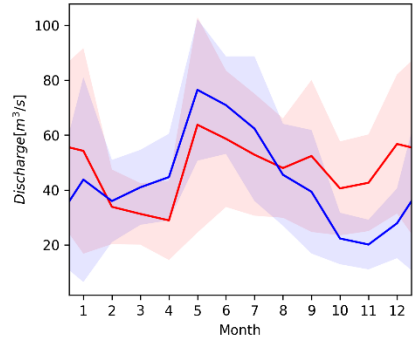
(a) Daily discharge - calibration period



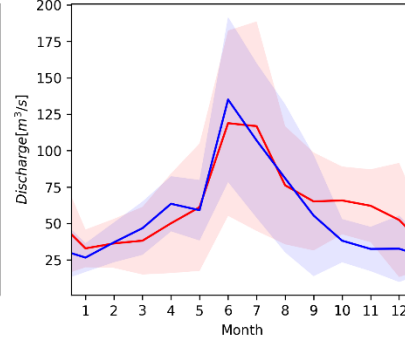
(b) Daily discharge - validation period



(c) Monthly discharge - calibration

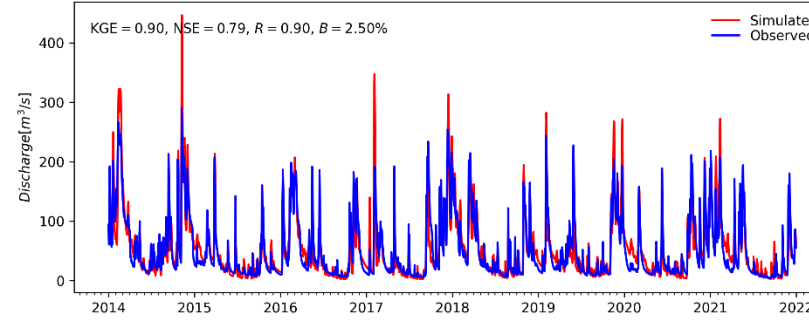


(d) Monthly discharge - validation

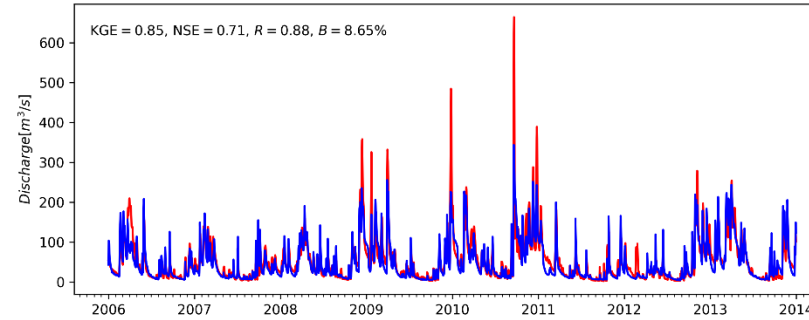


Ljubljana at Moste I, basin area: 1778 km²

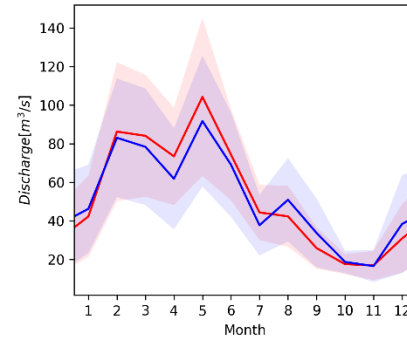
(a) Daily discharge - calibration period



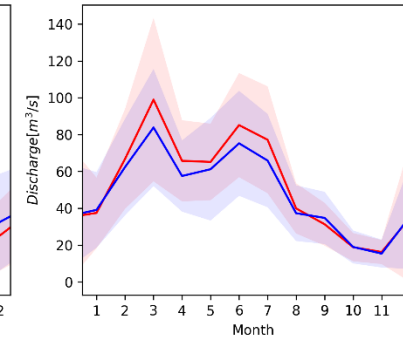
(b) Daily discharge - validation period



(c) Monthly discharge - calibration

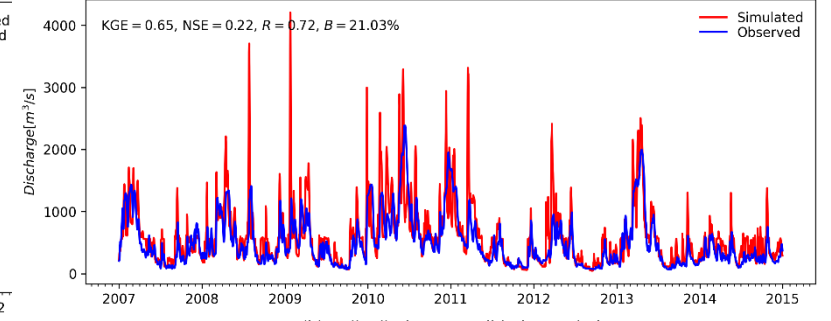


(d) Monthly discharge - validation

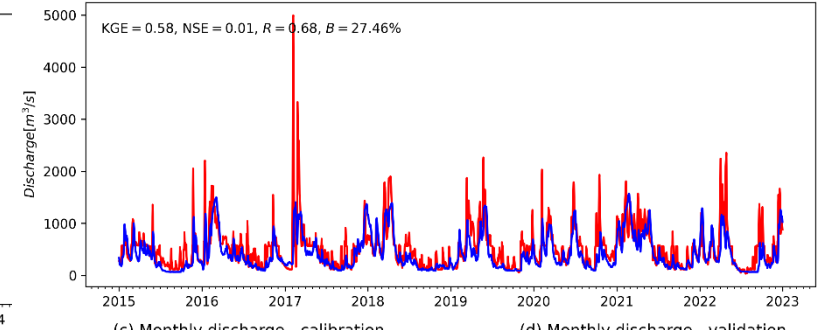


- at Szolnok, basin area: 73113 km²

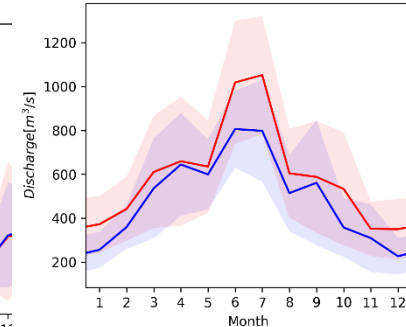
(a) Daily discharge - calibration period



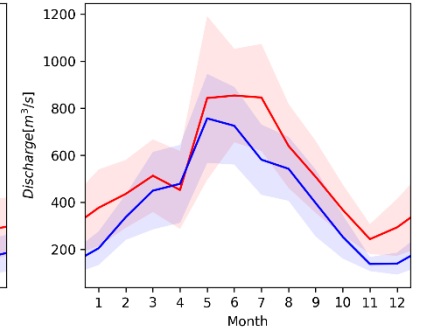
(b) Daily discharge - validation period



(c) Monthly discharge - calibration

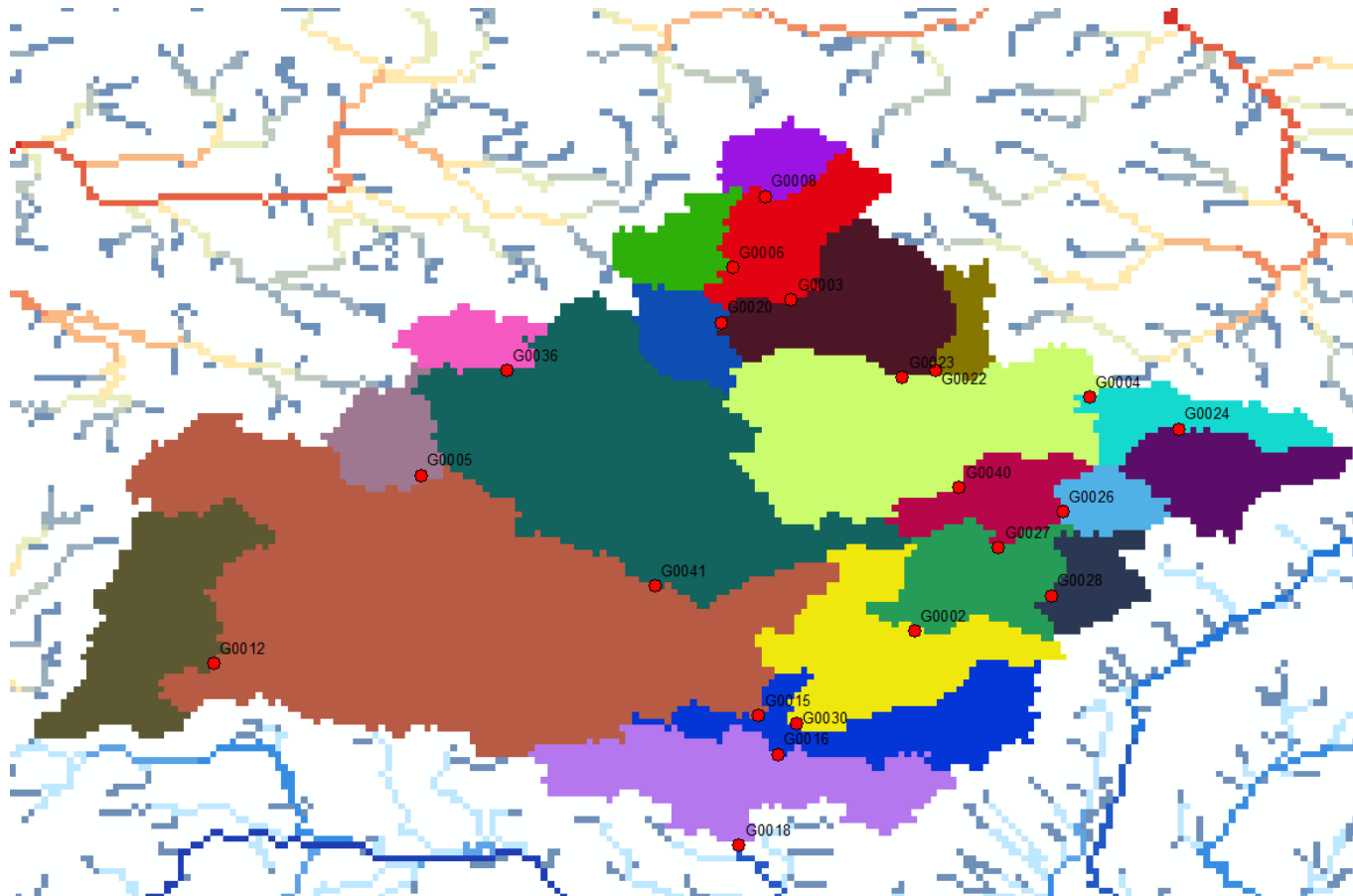


(d) Monthly discharge - validation



Calibration

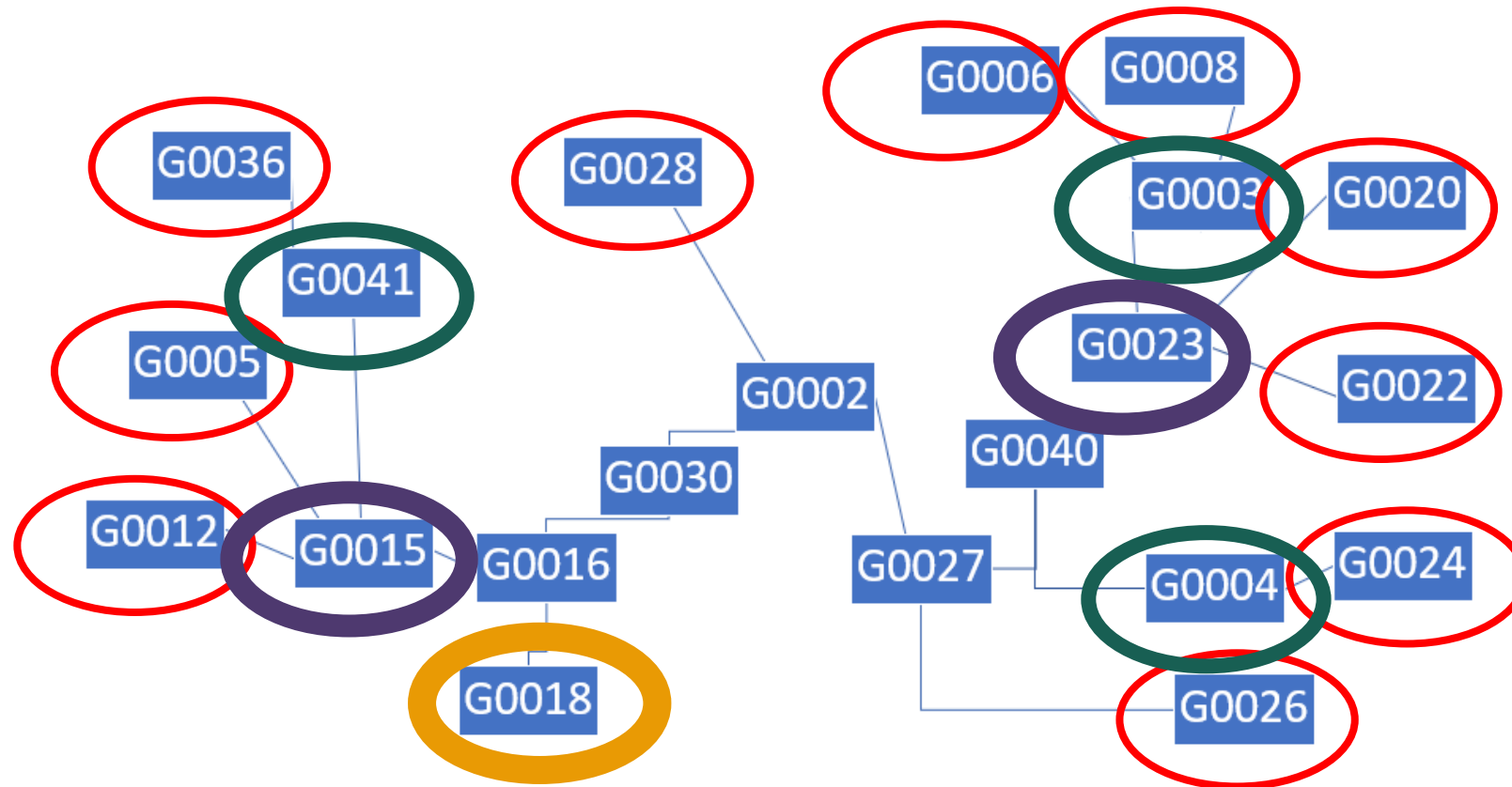
Upstream to downstream calibration



Morava basin with 21 subbasins

Calibration

Upstream to downstream calibration



Morava basin with 21 subbasins

Calibration

1. Running a fast calibration for an upstream basin

Go to your basin folder and /basin_calibration_tool

Correcting path in the settings files:

1) Settings1_fast.txt

Set the path name to your computer settings

[DEFAULT]

```
Root = C:/work/CWATM/Morava
RootLinux = /p/luc/watmodel/calibration/danube_1min
Rootbasin = morava_calibration_tool

# start date of simulation outputs
ForcingStart = 1/1/1990
.....
```

```
44 No_of_calibration_lists = 1
45 RunCwatm = python C:/work/CWATM/Morava/morava_calibration_tool/run_cwatm.py
46 RunCalib = python C:/work/CWATM/Morava/morava_calibration_tool/calibration_sing
47 listPC = listPC
48
```

Set this to: use_multiprocessing = 0

```
60 # if more than one node is to be used for parallel processing
61 use_multiprocessing = 1
62
```

Calibration

1. Running a fast calibration for an upstream basin

Go to your basin folder and /basin_calibration_tool/templates

Correcting path in the settings files:

2) settings_morava_1min2.ini

Set the path name to your computer settings

```

115 #-----
116 [FILE_PATHS]
117 #-----
118 PathRoot = C:/work/CWATM/Morava
119
120 PathOut = $(PathRoot)/out1
121 PathInit = $(PathRoot)/init
122 PathMaps = $(PathRoot)/input
123 PathMeteo = $(PathRoot)/meteo/emo-1
124
125 PathParameter = $(PathRoot)/parametermaps
126 Excel_settings_file = $(PathMaps)/landcover/crops/Crop_List_Danube_MAPSPAM2020.xlsx
127
128 #-----
129 [NETCDF_ATTRIBUTES]
130 institution = IIASA
131 title = Danube Water Model - EMO-1 Morava
132 metaNetcdfFile = C:/work/CWatM/Morava/morava_calibration_tool/cwatm/metaNetcdf.xml
133

```

Calibration

2. Running a fast calibration for an upstream basin

Go to your basin folder and /basin_calibration_tool/

Running a fast calibration

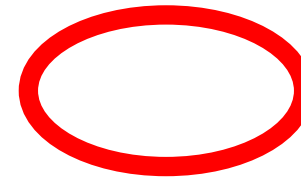
Execute: [runsingale.bat](#) or [python calibration_single_single.py](#)

This will launch a fast calibration in an upstream basin (here: G0022)

```
P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava\morava_calibration_tool
Start calibration
CWATM - Community Water Model 1.5 Date: 01.06.2021
International Institute of Applied Systems Analysis (IIASA)
Running under platform: Windows
-----

CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:\watmodel\CW
nd in P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava\morava_calibratio
ava_1min2-Run00_000.ini

Step      Date      Discharge
4748      31/12/2022      0.79
Start: 10:21:55.163750
End: 10:25:44.371689
run_rand_id: 00_000, KGE: 0.357, COR 0.453, NSE: 0.108
CWATM - Community Water Model 1.5 Date: 01.06.2021
International Institute of Applied Systems Analysis (IIASA)
```



Calibration

2. Running a fast calibration for an upstream basin

The fast calibration is using only:

- Initial population of 4 (normally ≥ 256)
- 2 generations (normally ≥ 10)
- 2 runs per generation (normally ≥ 32)

Please stop the run

Change back:

Use_multiprocessing = 1

→ the calibration is splitting the runs for multiprocessing

Run again: [runsinglе.bat](#)

Settings1_fast.txt – last part

```
[DEAP]
maximize = True

# if more than one node is to be used for parallel processing
use_multiprocessing = 0

# Number of nodes used for multiprocessing, recommended 1-8
pool_limit = 32

# Number of generations, recommended 10-30
ngen = 10
ngen = 2

# Initial population, recommended 128-256
#mu = 256
mu = 4

# Runs per generation, recommended 8-32
#lambda_ = 32
lambda_ = 2

# Select the best out of the generation
#select_best = 10
select_best = 2
```

```
P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava\morava_calibration_tool>python calibration_single_single.py
12 32
12
run_rand_id: 00_000, KGE: 0.357, COR 0.453, NSE: 0.108
```


Calibration

2. Running a fast calibration for an upstream basin

The fast calibration is using only:

- Initial population of 4 (normally ≥ 256)
- 2 generations (normally ≥ 10)
- 2 runs per generation (normally ≥ 32)

Please stop the run

Change back:

Use_multiprocessing = 1

➔ the calibration is splitting the runs for multiprocessing

Run again: [runsingl.e.bat](#)

Folder structure in:

morava_calibration_tool\calibration1\G0022\out
at start:

Name
00_000
00_002
00_003
00_004

Calibration

3. Look at the results

Your DOS box should look like this

```
run_rand_id: 00_002, KGE: 0.469, COR 0.571, NSE: 0.203
run_rand_id: 00_004, KGE: 0.529, COR 0.571, NSE: 0.210
>> gen: 0, KGE: 0.529
run_rand_id: 01_001, KGE: 0.552, COR 0.578, NSE: 0.177
run_rand_id: 01_002, KGE: 0.509, COR 0.554, NSE: 0.249
>> gen: 1, KGE: 0.552
run_rand_id: 02_002, KGE: 0.469, COR 0.571, NSE: 0.260
run_rand_id: 02_001, KGE: 0.553, COR 0.585, NSE: 0.204
>> gen: 2, KGE: 0.553
>> Termination criterion ngen fulfilled.
>> Saving optimization history (front_history.csv)
>> Saving Pareto optimal solutions (pareto_front.csv)
>> Running Model using the "best" parameter set
CWATM - Community Water Model 1.5 Date: 01.06.2021
International Institute of Applied Systems Analysis (IIASA)
Running under platform: Windows
-----

CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava
and in P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava\morava_calibration_tool\calibration1\G0022\out\03
rava_1min2-Run03_best.ini

Step      Date      Discharge
12053  31/12/2022      1.73
Start: 10:42:24.688515
End:    10:50:39.475264
>> Saving "best" simulated streamflow (streamflow_simulated_best.tss)
=====DONE == G0022 =====
```

Folder structure in:

morava_calibration_tool\calibration1\G0022\out
at end of the calibration:

-  inflow
-  maps
-  out
-  checkpoint.pkl
-  front_history.csv
-  pareto_front.csv
-  runs_log.csv
-  streamflow_simulated_best.tss

After the calibration, the subbasin is run again for a greater time period

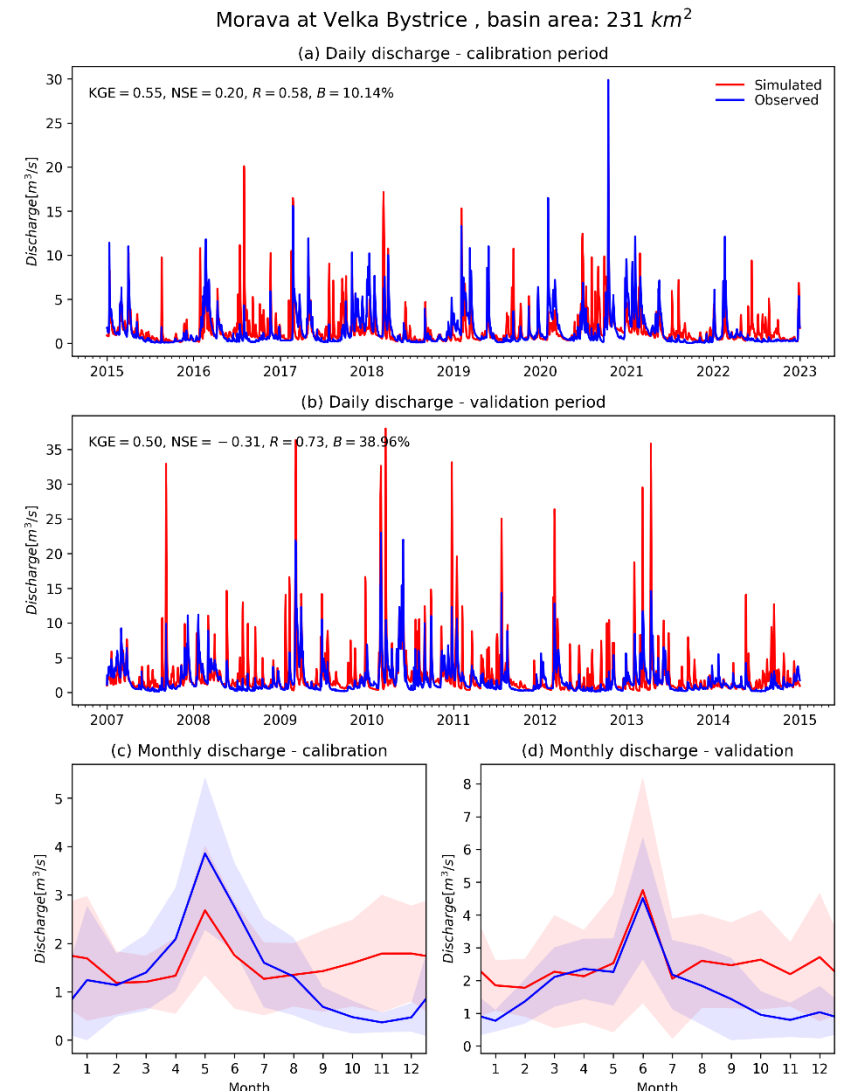
Calibration

3. Look at the results

Execute: `runC10.bat` or
`python CAL_10_COMPUTE_STATS_AND_FIGURES_TIME_SERIES.py settings1_fast.txt`

To create a figure in:
`yourbasin_calibration_tool\calibration1\FIGURES`

This is the single basin calibration:
 For a “real calibration” you have to change the [DEAP]
 parameter in `settings1_fast`

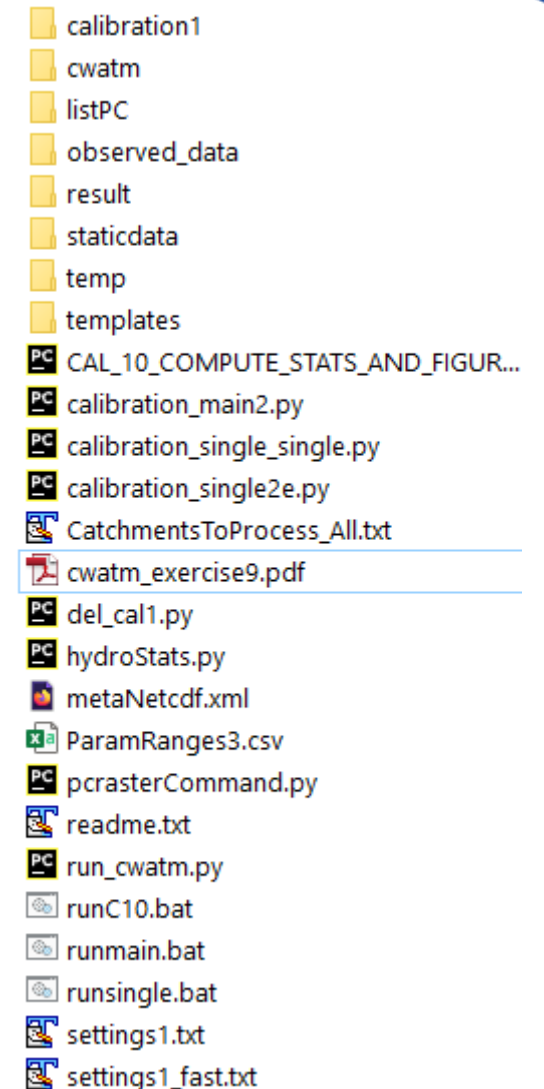


G0022_Morava_Velka_Bystrice_.png

Calibration

4. Some more information

- Calibration1: has the results for different subbasin
in the beginning it should have only inflow, maps, out
- cwatm: A copy of the model
- ListPC: This has a list of subbasin. If you run top to downstream
calibration with several computer:
it tells each computer what is done, what is running
- Observed_data: Has the metadata and the discharge timeseries
of subbasin stations
- Result,staticdata,temp: Some folders to store some inbetween data
- Templates: hold the settingsfile for calibration with placeholders:
e.g. SnowMeltCoef = %snowmelt
Placeholders are filled in during each calibration run



Calibration

4. Some more information

CAL_10_COMPUTE_STATS_AND_FIGURES_TIME_SERIES.py:

Tool to produce figures for each subbasin

calibration_main2.py: Prg to run a upstream to downstream calibration

calibration_single2e.py: Prg which is executed by calibration_main2.py for each subbasin. First the upstreams then working down

calibration_single_single.py: Prg which runs a single basin. No of subbasin is hard coded

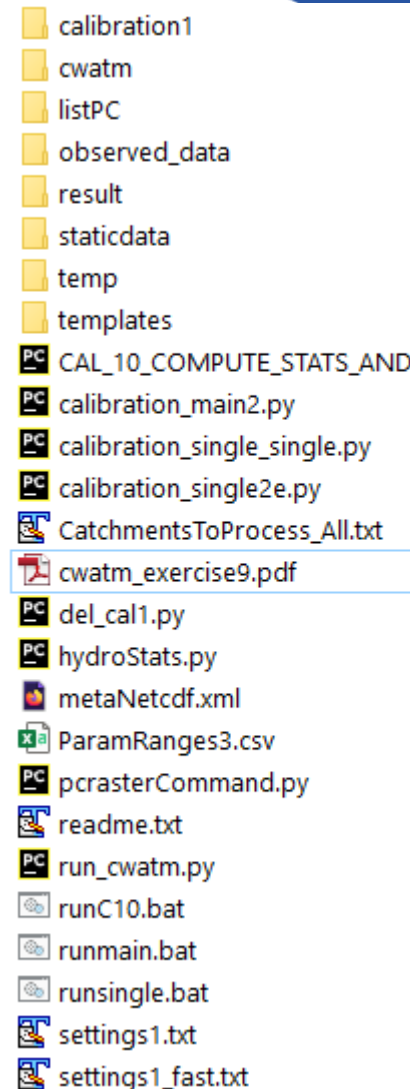
del_cal1.py: Deletes all runs in calibration1

hydroStats.py Has all the objective functions

ParamRanges3.csv Here the parameter and thweir range are stored

settings1_fast.txt: Settings_file for a single subbasin

settings1.txt: Settingsfile for a top-down calibration



Calibration

What can go wrong 1

Run runsingle.bat

`python calibration_single_single.py`

- Your python version is starting with python3 or you have to put in the full path of your python version
c:/python38/python3 calibration_single_single.py
- You did not change the path in settings file settings1_fast.txt and templates/settings_morava_1min2.ini
- You did not download the meteo data from Sharepoint or you did not download other data
- You still have problems with the gdal library
 - use the newest version of CWatM on Sharepoint
 - install gdal like in <https://cwatm.iiasa.ac.at/setup.html#external-libraries>
- You have problems with white space in file names
Use e.g. "c:/white space/python.exe" calibration_single_single.py



Calibration

What can go wrong 2

Run runsingle.bat

`python calibration_single_single.py`

- You did not add the libraries deap, matplotlib and pandas
- Some results are already in basin/calibrationtool/calibration1/G0022 (or the G00?? You are using)
- delete the files and the /out directory, make a new out directory
- You cannot load the settings files,
Your settings file are not stored as utf-8.



Calibration

1. What can go wrong 3

Run runsingle.bat

`python calibration_single_single.py`

You run it but it crash:

- Go to basin_calibration_tool/calibration1/G00??/out
- Is there a folder 00_000?
 - No: bad, go back to what can go wrong 2
 - yes: good, go into 00_000
- Inside 00_000 there is a file runpy00_000.bat
 - type cmd in the explorer to start a new DOS box
 - type runpy00_000.bat and execute

CWatm is running?

Yes: very good – check if the observed data in

basin_calibration_tool \observed_data/ qgis_yourbasin_5.csv are in right format

No: change settings_fast-Run00_000.ini (e.g. path name) till CWatM is running

copy the changes into templates/settings.ini

Please have a look at:

<https://cwatm.iiasa.ac.at/calibration.html>

https://cwatm.iiasa.ac.at/calibration_tutorial.html